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Supplement Use in Patients Undergoing Anterior Cruciate Ligament Reconstruction: A Systematic Review

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Abstract

Purpose

The purpose of this systematic review is to assess whether a standardized dietary supplementation can help to decrease postoperative muscle atrophy and/or improve rehabilitation outcomes in patients who underwent anterior cruciate ligament reconstruction (ACL-R).

Methods

A systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA). MEDLINE, Scopus, and Cochrane Library databases were searched, and articles that examined protein or amino acid, vitamin, or any other type of supplementation in ACL-R were reviewed. Two independent reviewers conducted the search utilizing pertinent Boolean operations.

Results

A total of 1,818 articles were yielded after our database search. Ten studies fulfilled our inclusion criteria and only assessed patients undergoing ACL-R. Four studies assessed protein-based supplementation. One study assessed creatine as a supplement. Four studies assessed vitamin-based supplementation. One study assessed testosterone supplementation.

Protein and amino acid supplementation showed potential benefits; multiple authors demonstrated a combination of improved achievement of rehabilitation benchmarks, graft maturation, muscular hypertrophic response, and peak dynamic muscle strength. When looking at creatine, vitamin, or hormone-based protocols, none demonstrated results suggesting these factors may attenuate muscle atrophy after surgery. Vitamin C and E demonstrated potentially increased local inflammation in skeletal

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muscle, which runs contrary to the belief that antioxidant vitamin-based supplementation may decrease the inflammatory response that plays a role in the post injury/operative period.

Conclusion

Protein-based supplementation may play a role in mitigating muscle atrophy associated with ACL-R, as multiple authors demonstrated a combination of improved achievement of rehabilitation benchmarks, thigh hypertrophic response, and peak dynamic muscle strength. However, based on current literature, it is not possible to recommend a specific protein-based supplementation protocol at this time for patients undergoing ACL-R. Limited evidence suggests no benefit for creatine, vitamin, or hormone-based protocols.

Level of Evidence: II, a systematic review of level I-II studies

Keywords

Supplement; orthopedics; surgery; knee; muscle mass; atrophy; ACL; Anterior Cruciate Ligament

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Introduction

Post-operative muscle atrophy is a common sequela following orthopedic surgery and has a multitude of causes, including pre-operative diminished activity secondary to pain, the presence of inflammation, nutritional deficiencies, and limited muscle activation secondary to musculoskeletal injury [1, 2]. The post-operative catabolic inflammatory state and subsequent immobilization and neuromuscular activation defects further contribute to post-operative muscle loss and residual strength impairment [3-5]. For orthopedic procedures involving the knee, such as anterior cruciate ligament reconstruction (ACL-R), this process is significant because the loss of quadriceps muscle strength/size serves as a major barrier to post-operative physical rehabilitation and return to sport [6].

The task of reducing post-operative quadriceps muscle atrophy and accelerating the return of quadriceps muscle strength seen in ACL-R patients has been studied in relation to multiple parameters, with increased focus on post-operative rehabilitation. For example, some physical rehabilitation protocols now incorporate electrical stimulation or immediate weight bearing with early removal of assisted walking devices as these measures are believed to accelerate return of muscle mass. [4]. However, despite these efforts patients continue to have reduced muscle strength and functional performance postoperatively [7]. Therefore, further optimization in post-operative strategies targeting improved return to function in ACL-R patients is possible and warranted.

Research has also been performed evaluating the addition of perioperative nutrition supplementation and its effects on postoperative muscle atrophy [8]. It has been proposed that protein, peptide, or essential amino acid supplementation may not only help to maintain adequate protein synthesis levels in skeletal muscle and restore any nitrogen balance deficiencies following orthopedic trauma, but may also serve to increase the capacity to synthesize protein and repair muscle tissue postoperatively, which has already been shown in patients undergoing total knee arthroplasty [8, 9]. There have also been studies regarding

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vitamin supplementation with supplements touted as antioxidants in order to reduce the proinflammatory state seen after injury or surgery. While this, in theory, seems beneficial in the postoperative state, these studies have linked these supplements with impairments in muscle strength and subsequent muscle atrophy [10, 11]. This highlights the importance of evidence-based practice, not only with regards to surgery and rehabilitation, but also with regards to diet.

While prior studies have demonstrated the benefits of protein supplementation in an arthroplasty population, there is a paucity of literature indicating the efficacy of these interventions in a younger more active cohort such as patients undergoing ACL-R surgery. Given that current approaches may still be insufficient in helping ACL-R patients restore pre-injury muscle strength and function, there may be value in assessing perioperative supplementation as an additional modality for ACL-R rehabilitation. Therefore, the purpose of this systematic review is to assess whether a standardized dietary supplementation can help to decrease postoperative muscle atrophy and/or improve rehabilitation outcomes in patients who underwent anterior cruciate ligament reconstruction (ACL-R). Our hypothesis is that supplementation in this population may not be necessary compared to older, sarcopenic patients undergoing orthopedic procedures.

Methods

Search Strategy

Two independent reviewers (D.G and C.E) performed a systematic review of MEDLINE, The Cochrane Library, and Scopus databases according to the Preferred Reporting Items for Systematic Reviews and Meta Analyses (PRISMA) guidelines using a PRISMA checklist. The goal of this review was to identify all studies related to the use of proteins, amino acids, vitamins, and other nutrition supplements commonly used to treat postoperative muscle atrophy after ACL-R. Two reviewers independently conducted literature searches in July of 2019 using PubMed (MEDLINE), Scopus, and Cochrane

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Library databases. Each search included a combination of the following terms and Boolean operations: ("amino acid supplement" or "protein supplement" or "vitamin supplement" or "hormone supplement " or "amino acid additive" or "protein additive" or "vitamin additive" or "hormone additive" AND ("orthopedics" or "surgery" or "sports" or "knee" or "ACL" or "anterior cruciate ligament"). Though our primary focus was ACL-R, some search terms were intentionally left broad to see if there was literature addressing other sports related arthroscopic procedures of the knee. Our intent of this procedure is to be able to comment on broader supplementation use in sports related procedures of the knee outside of ACL-R. A fellowship-trained knee surgeon (M.G.B) reviewed all discrepancies in inclusion or exclusion of studies.

Inclusion and Exclusion Criteria

The main criteria for selection were articles within the aforementioned databases that were randomized control trials (RCT), written in English, and utilized protein, vitamin, hormone, or other supplementations to mitigate muscle atrophy or improve rehabilitation outcomes following ACL-R. Articles outside this scope were excluded. Both reviewers independently filtered through all returned articles based on the title and abstract, identified appropriate articles, and any subsequent discrepancies were co-reviewed. Each study's reference list was also reviewed for additional articles.

Our analysis of each article included the following parameters: demographic data of participants (age, gender, BMI), total number of participants, supplement combination used, the length of the study, potential complications, and outcomes. Primary outcomes of interest include functional scores assessing muscular strength or thigh muscle measurements assessing post-operative atrophy. Secondary outcomes of interest include but are not limited to pain levels, analgesic use, or return to sport or work.

Data Analysis

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Two independent reviewers extracted the data (D.G and W.H.C) with any discrepancies reviewed by the senior author (M.G.B). The risk of bias was then assessed using the Cochrane Risk of Bias tool for Randomized Controlled Trials, which evaluates for seven different types of bias. Based on specific criteria, all studies were then classified as “low risk”, “high risk”, or “unclear risk” for each type of bias . The results of such assessments were then converted to Agency for Healthcare Research and Quality standards in order to ultimately rank each RCT as “good quality”, “fair quality”, and “poor quality”. Level of evidence of each article was re-evaluated using criteria established by Hohmann et al [12].

Statistical Analysis

A statistician reviewed the methodology, treatment groups, and reported outcomes of the included studies in order to evaluate the possibility for further meta-analysis of each treatment group. All statistics were performed on SAS 9.4 (SAS Institute).

Results

Literature Search

Dual review of current, relevant literature resulted in a total of 1,818 articles for review, of which 1,181 were kept for screening after duplicates were removed (Figure 1). Both reviewers finalized a list of 37 articles based on title and abstract for further full-text screening. Twenty-seven manuscripts regarding orthopedic surgery and amino acid or vitamin supplementation were excluded as they did not pertain to ACL-R or met our inclusion criteria. Of note, no manuscripts addressing sports related arthroscopic procedures of the knee outside of ACL-R were found.

Study Characteristics

Ten studies with 292 patients were included. All studies were prospective RCTs published between 2004-2019. All of these studies focused on patients who had undergone or planned to undergo ACL-R and were pre or post-operatively given supplements. Of note, no studies exploring supplements after

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other arthroscopic/sports procedures were found. The study characteristics are illustrated in Table 1 while quality and risk of bias are both shown in Table 2.

Protein Based Supplementation

Four RCTs including 177 patients (142 male, 35 female) evaluated the use of protein or amino acid-based supplementation, three of which administered supplements after ACL reconstruction and one prior to surgery [13]. All four studies assessed a combination of rehabilitation benchmarks and isokinetic testing such as peak dynamic strength, two of which either found a significant difference in isokinetic metrics ($P < 0.01$) or number of physical rehabilitation sessions ($P < 0.001$) in order to match pre-determined benchmarks favoring protein-based supplementation compared to a placebo [13, 14]. Two studies assessed International Knee Documentation Committee (IKDC) scores and a combination of analgesia consumption and visual analogue scale (VAS scores,) with Vidriero et al. concluding that the protein-based group had both significantly improved IKDC scores ($P = 0.029$) and reduced analgesia consumption (50% vs 8.5% respectively, $P < 0.05$) [14, 15]. Two studies assessed thigh perimeter as a measure for muscle hypertrophy after surgery, and both found a significant increase in thigh hypertrophy in the protein-based group compared to placebo ($P < 0.01$ and $P = 0.009$ respectively), though Holm et al's muscle biopsy analysis demonstrated no difference between their protein and placebo [13, 16]. Only one study assessed graft maturation via magnetic resonance imaging, concluding that patients with protein supplementation demonstrated more advanced graft maturation (grades three and four) at final follow-up (61.7% of supplement group versus 38.4% control group) [14]. However, it must be noted that the significance of this finding remains unclear, since Vidriero et al.'s study reported two P values differing in statistical significance ($P < 0.01$, $P= 0.05$) and thus warrants further review [14].

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Creatine Based Supplementation

Only one RCT including 60 patients (33 male, 27 female) evaluated creatine as a potential supplement to curb muscle atrophy and improve knee function after ACL reconstruction [17]. The authors measured Knee Outcome Survey responses, body fat, goniometric knee goniometric knee range of motion, isokinetic knee strength/power and hip strength up to 24 weeks after surgery. The authors found that there were no significant differences in any of the aforementioned metrics between the creatine and placebo groups. Of note, four patients dropped out of the study due to gastrointestinal distress associated with the supplement, the only reported side effect from any supplement intake amongst all the studies in this systematic review [17].

Vitamin Based Supplementation:

Four RCTs by Barker et al. included 49 patients (36 male, 13 female) in order to assess the feasibility of vitamin supplementation in the context of ACL reconstruction, in particular assessing how both vitamin C and E may modulate the pro-inflammatory state as a result of surgery. Using the same patient group, Barker et al. drew blood samples in order to assess antioxidant levels and circulating cytokines and determined that by 72 hours serum inflammatory cytokine levels were similar between both vitamin and placebo groups [18-20]. The authors also assessed muscle power, which was similar between both groups, and peak isometric force, which was greater in the placebo group after three months. However, muscle biopsies in the vitamin group demonstrated significantly increased infiltration of inflammatory cells after surgery compared to placebo ($P < 0.05$).

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In 2015, Barker et al. also performed a similar study with the addition of the gamma form of vitamin E to their supplementation group and the assessment of IKDC and Western Ontario and McMaster University Osteoarthritis Index (WOMAC) scores while forgoing a muscle biopsy [21]. The authors also concluded that despite the addition of another form of vitamin E, supplementation with both vitamin E or C was ineffective at modulating serum cytokine levels or improving muscle strength and function after both ACL injury and surgery.

Hormone Based Supplementation:

One RCT including six male patients assessed the effects of hormone supplementation on muscle mass after ACL-R via intramuscular injection of 200 mg testosterone weekly from two weeks prior to surgery to six weeks after surgery [22]. The primary outcome was change in total lean body mass from baseline while secondary outcomes included maximal extensor muscle strength and Knee Injury and Osteoarthritis Outcome Scores (KOOS), with a maximum follow-up of 24 weeks. The authors found that total lean body mass was significantly greater in the testosterone group compared to the placebo group (P = 0.01) at six weeks, but at no other point was there a statistically significant difference between both groups. The authors also found that extensor strength in the uninjured leg significantly favored the testosterone group at 12 weeks postoperatively (P=0.04), but again, statistical significance was not seen at any other time point. Most notably, there were no differences between both groups regarding extensor strength of the injured leg or clinical scores at any point including 24 weeks after surgery. No adverse effects of testosterone were reported throughout the duration of the study.

Discussion

The most important finding in our study was that only protein or amino acid supplementation in the context of ACL-R may play a role in mitigating the muscle atrophy associated with surgery as multiple authors demonstrated a combination of improved achievement of rehabilitation benchmarks, thigh

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hypertrophic response, and peak dynamic muscle strength. Creatine, vitamin, or hormone supplementation were not shown to improve post-operative outcomes in ACL patients, rather vitamin use may have instead been detrimental to muscle recovery after ACL-R.

The goal of this systematic review was to evaluate if a standard supplementation protocol may play a role in improving post-operative outcomes for patients undergoing ACL-R. The rationale for examination of supplementation use for ACL-R is tied to the fact that protein supplementation use has already been studied extensively in relation to total knee arthroplasty and has demonstrated to preserve lean body mass, maintain functional ability, and attenuate muscle atrophy postoperatively [8, 9]. However, the study of supplementation as it relates to ACL-R, let alone other arthroscopic/minimally invasive orthopedic procedures involving the knee, nonetheless remains limited since current research still remains primarily focused on patient populations that are at higher risk of nutritional deficiencies and subsequent sarcopenia such as those undergoing TKA, while patients undergoing sports related procedures involving the knee tend to be younger and more nutritionally well balanced [8, 23]. Furthermore, based on our broad Boolean search no study addressed other sports related arthroscopic procedures of the knee outside of ACL-R, thereby limiting the generalizability of our findings to other sports related orthopedic procedures of the knee. Even within the ACL literature, different ACL graft types have also been shown to influence on muscle atrophy, which no study has addressed to date [24, 25].

That being said, protein supplementation demonstrated the most promising results compared to the other supplementation types and warrants further study. Vidriero et al. demonstrated the most promising results, as their supplement protocol led to decreased time needed to achieve appropriate rehabilitation benchmarks, reduced analgesic consumption, improved IKDC scores, and more advanced graft maturation [14]. However, a few limitations exist, the most significant being that their supplement

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Progen is a combination of plasma proteins, hydrolyzed collagen, hyaluronic acid, glucosamine sulfate, and vitamin C, as their rationale was to provide substances that can enhance the healing of the ACL graft, reduce pain, and decrease proinflammatory cytokines [26, 27].

Two other protein studies also found promising results, showing significant differences in hypertrophic response and isokinetic strength testing favoring the supplementation group [13, 16]. Holm et al. in particular found that protein supplementation led to a significant increase in hypertrophic response and peak dynamic muscle strength in the distal quadriceps, particularly in the vastus medialis [13]. Their unique analysis of the vastus medialis may have important ramifications, as this muscle both atrophies and recovers at an accelerated rate compared to the other quadricep muscles, suggesting a high turnover rate that makes the muscle itself more susceptible to changes in amino acid and protein balance after a stressful insult compared to muscles with lower turnover rates [28-30]. These points all suggest that protein supplementation may play an important role in correcting the muscle atrophy seen in ACL reconstruction by maintaining a positive protein balance in high turnover muscle.

Interestingly, Eraslan and Ulkar opted to use Glucosamine which is a protein based product with analgesic and anti-inflammatory properties, which the authors believed may assist patients during the acute rehabilitation period where persistent knee pain may further reduce physical therapy compliance and thus accelerate muscle atrophy [15, 31]. However, the authors found no benefits with Glucosamine supplementation, which was controversial compared to previous literature on Glucosamine which demonstrated improved functional capacity in patients with osteoarthritis of the knee [32-34]. That being said, the main flaw of this study was the addition of supplementation six weeks after the patient's surgery rather than immediately post-operation, where pain, inflammation, and muscle atrophy is greatest, thereby making it difficult to discredit Glucosamine as an adjunctive therapy entirely for patients undergoing ACL reconstruction [35].

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Only one study assessed creatine as a potential supplement because creatine has been postulated to enhance one's ability to maintain muscular force and power during strength training by increasing muscular stores of creatine phosphate, thus maximizing the potential of post-operative strengthening exercises [17, 36]. In fact, Hespel et al. further contributed to this rationale when they demonstrated that creatine supplementation can assist in strength training for subjects following limb immobilization, which may simulate the immediate post-operative period for ACL patients [37]. However, the authors found that by six months isokinetic muscle strength and power in the knee, knee outcome survey scores, and single leg hop testing were not significantly different between both cohorts, nor were there significant differences in hip flexion, abduction, or adduction either. These findings later corroborated another limb immobilization study that concluded creatine supplementation prior to and during leg immobilization does not prevent or attenuate the loss of muscle mass [38]. This may be due to creatine being an energy producing substrate for muscle, and without the necessary amount of protein required to permit hypertrophy of atrophied muscle, increasing the amount of energy stored in muscle may have limited effect on muscle strength and size. Of note, four patients did drop from Tyler et al's study due to gastrointestinal distress associated with the supplement, the only reported side effect from any supplement intake amongst all the studies in this literature review. Though creatine use is associated with gastrointestinal distress, a recent meta-analysis concluded that creatine supplementation in moderation is safe for healthy individuals, but excessive use has been associated with renal impairment [39].

Only one study group has addressed vitamin supplementation, believing that the antioxidant properties of both vitamin C and E may blunt the pro-inflammatory state caused by injury and surgery and subsequently reduce associated muscle atrophy [40, 41]. However, instead the authors found no significant differences in both serum inflammatory cytokine levels within 72 hours of surgery and thigh

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circumference, and the placebo group displayed increased peak muscle power over the vitamin group at three months [18-20]. Furthermore, muscle biopsies demonstrated supplementation significantly promoting the infiltration of inflammatory cells in skeletal muscle, suggesting vitamin E and C may instead promote certain local inflammatory processes within skeletal muscle during early recovery. In 2015, Barker et al. concluded that despite the addition of the gamma form of vitamin E, vitamin antioxidant supplementation was ineffective at modulating serum cytokine levels or improving muscle strength after ACL-R [21]. These findings suggest that antioxidant vitamins in a healthy population may in fact potentially hinder strength recovery of ACL injured limbs three months post-surgery, though the outcome of other vitamin supplementations remains unknown.

Surprisingly, only one study assessed hormone supplementation in the form of testosterone in the context of mitigating muscle atrophy after ACL-R or any other sports related orthopedic procedure of the knee. Wu et al. hypothesized that testosterone may play a role in the perioperative recovery of ACL patients based on previous literature suggesting testosterone administration increases lean muscle mass (which correlates with muscle strength), but no previous studies have shown that testosterone plays a role in the surgical recovery period by offsetting loss of leg strength due to surgery and subsequent immobility [22, 42]. Though Wu et al. found a significant difference in total lean body mass and extensor strength in the uninjured leg at one follow-up period, by 24 weeks testosterone use did not have any beneficial effects, particularly in the injured leg, which the authors attributed to their study not being sufficiently statistically powered to detect differences in strength or clinical outcomes. While no adverse effects were reported, the authors noted that testosterone administration is nonetheless limited to low dosages for male eugonadal patients due to testosterone being a potent hormone acting on various organ systems, which is especially true for female patients hence their exclusion from the study. Based on the above, the lack of benefit regarding testosterone administration to the injured leg post-operatively

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coupled with the safety risk of consistent testosterone administration, which can only be administered to a specific population for a short period of time, does not support the use of testosterone for mitigating of post-operative muscle atrophy at this time.

Though there remains minimal literature assessing the effects of hormones on post-operative muscle recovery in sports related orthopedic procedures of the knee, there is literature addressing hormone use for other pertinent post-surgical outcomes. For example, a Rue et al's case series assessed the effects of hormones in the form of oral corticosteroids on such procedures, concluding that for ACL patients oral corticosteroids may assist in successful return to normal range of motion [43]. Multiple studies have also concluded that corticosteroids may play a role post-operative pain and inflammation modulation within the knee, opting to even suggest intra-articular injections after surgery [44, 45]. However, to our knowledge, no studies have explored the potential role for Human Growth Hormone in post-operative muscle recovery for such procedures. Overall, future RCTs should assess whether testosterone, corticosteroids, or HGH can safely improve post-operative recovery in patients undergoing sports related orthopedic procedures of the knee.

Limitations:

This present study has similar limitations, the first being the limited number of studies that were eligible for review due to the strict inclusion criteria, as multiple studies were excluded from this review due to the type of orthopedic procedure performed, namely joint arthroplasties in order to focus on a younger, more active patient population compared to older populations with higher baseline risk of sarcopenia and vitamin/or protein deficiency [8, 23, 46]. Though our primary objective was to address supplementation in ACL-R, our broad search terminology did not capture any additional articles addressing supplementation in other sports related arthroscopic procedures. Therefore, we cannot comment on supplementation use in other sports related arthroscopic procedures of the knee outside of

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ACL-R. Furthermore, one research group used the same patient cohort for three different studies though the primary outcomes for each study were different, thus preventing a meta-analysis from being performed for vitamin-based supplementation.

Though our focus on only RCTs further reduced the number of applicable studies but given the number of variables that must be controlled in a study examining supplement intake such as supplement dosage, timing of supplement intake, general dietary intake, or rehabilitation regiment, the authors concluded that only an RCT can best limit as many confounding variables as possible. Despite this focus, multiple studies, particularly the protein supplementation group, nonetheless varied in study design, patient demographics, supplementation dosage, type and timing, rehabilitation regiments, patient follow-up, standards for diet control, and outcome measures of interest, thus leading to small study populations for each class of supplement. These variation made each study too clinically diverse and thus prevented limited more in-depth statistical analysis comparing performance of different supplements, thereby preventing the authors from performing a statistically accurate and unbiased reliable meta-analysis.

Finally, this systematic review is ultimately limited by the level of evidence and quality of available literature (half of the studies were labeled as “poor”, and only one study met “good” quality), which prevents the authors of this review from making an evidence-based recommendation without the presence of bias, thus currently preventing clinicians from obtaining any substantial information that may change their current practice.

Conclusion

Protein-based supplementation may play a role in mitigating muscle atrophy associated with ACL-R, as multiple authors demonstrated a combination of improved achievement of rehabilitation benchmarks, thigh hypertrophic response, and peak dynamic muscle strength. However, based on current literature, it is not possible to recommend a specific protein-based supplementation protocol at this time for patients

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undergoing ACL-R. Limited evidence suggests no benefit for creatine, vitamin, or hormone-based protocols.

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Figure Legend

Figure 1: PRISMA Flow Chart

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Table 1: Summary of Included Studies

Author, Year	Study Design	# of Total Subjects	Gender (M//F)	Age	BMI	Endpoints	Before or After ACL-R	Treatment Group & Supplement Used	Control Group	Study Time	Summary of Outcomes	Level of Evidence
Holm, 2005 [13]	Double-Blinded RCT	26	16 M, 10 F	18-35	Protein + Carb (PC): 24.7+/- 1.1; Isocaloric-Carb (IC): 26.0 +/- 1.0; Placebo: 25.6 +/- 1.2	Isokinetic testing, quadriceps cross-sectional area, muscle biopsy	Before	Protein (n=8) : 10g protein, 7g carbohydrate, 3.3g fat; IC (n=9): 17g carbohydrate, 3.3g fat	Placebo (n=9): 1.4 g carbohydrate and 1 g fat	12 weeks	Protein group showed significantly larger hypertrophic response and peak dynamic muscle strength in the distal quadriceps compared to placebo (P < 0.01).	II
Vidriero, 2019 [14]	Non-Blinded RCT	72	60 M, 12 F	18-55	Experimental: 24.64 +/- 3.70 Placebo: 24.54 +/- 2.43	VAS, IKDC, Graft Maturation, Time of Achievement of Rehabilitation Benchmarks	After	(n= 34) (1 sachet daily). 2500 mg of CH, 300 mg of porcine PRP, 50 mg of HC-15, and 40 mg of vitamin C	(n=34) pain control and rehabilitation only for 90 days	90 days	IKDC scores were significantly higher in the supplement group (62.5 ± 11.7 vs 55.5 ± 11.1; P = .029) Patients in the control group needed more rehabilitation sessions than those in the supplemented group (mean of 48.4 ± 11.3 versus 38.0 ± 9.3 sessions, P < .001). Patients in the control group had elevated analgesic consumption compared to the supplement group after 30 days (50% vs 8.5% respectively, P < 0.05). More patients in the supplement group showed advanced degrees of graft maturation on MRI (P = .05)	II
Eraslan, 2014 [15]	Blind RCT	34 male athletes	34 Male	18-40	Not Provided	VAS, IKDC, LYS, Isokinetic Testing	After	(n=15) Glucosamine Sulfate 1000 mg daily	Placebo (n=15)	8 weeks	No significant differences were found regarding pain (VAS), functional status (IKDC, LYS) and muscular strength (isokinetic test) between the glucosamine and placebo groups .	II
Laboute, 2013 [16]	Double-Blinded RCT	45	32 M, 13 F	Mean: 24; Range: 18-45	Leucine: 25 +/- 3.3; Placebo: 25.5 +/- 2.9	Thigh perimeter, isokinetic testing, BMI	After	Leucine (n=22): 4 tablets per day (1.2 g leucine, 0.12 g carb; 0.01 g lipid, 6.59 kcal) 3x per day	Placebo (n=23); same sized tablet given at same intervals throughout day	2.7 weeks	At 10 cm from the patella, the perimeter of the injured side had undergone greater significant increase in the supplement group than in the placebo group (P = 0.009). All other parameters showed no significant improvement.	II
Tyler, 2004 [17]	Double-Blinded RCT	60	33 M, 27 F	mean: 36 M/37 F	Not Provided	Isokinetic testing, KOS, BMI	After	(n=23), Creatine 20g per day for first week, then 5g per day after	Placebo (n=28)	12 weeks	The results demonstrate that patients do not benefit from creatine supplementation during the first 12 weeks of rehabilitation after ACL reconstruction.	I
Barker, 2009 [18]	Double-Blinded RCT	20	20 Male	18-45	Experimental: 28.9 ± 2.1 Placebo: 29.7 ± 2.0	Isokinetic testing, muscle biopsy, blood samples	Before and After	(n=10); 200 IU Vitamin E and 500 mg Vitamin C; 2x day	Placebo (n=9): 313 mg soybean oil, 110 mg gelatin, 10 mg water, 7.5 mg beeswax, 8.5 mg carob, 5 mg lecithin	14 weeks	Antioxidant vitamin supplementation increased the infiltration of inflammatory cells and increased calpains 5 days after surgery (P < 0.05). The placebo group also demonstrated significantly improved peak isometric force at three months compared to the vitamin-based group (P < 0.05).	II

Barker, 2009 [19]	Double-Blinded RCT	20	20 Male	18-45	Experimental: 28.9 ± 2.1 Placebo: 29.7 ± 2.0	Isokinetic testing, muscle biopsy, blood samples	Before and After	(n=10); 200 IU Vitamin E and 500 mg Vitamin C; 2x day	Placebo (n=9): 313 mg soybean oil, 110 mg gelatin, 10 mg water, 7.5 mg beeswax, 8.5 mg carob, 5 mg lecithin	14 weeks	Supplementation with vitamins E and C prior to ACL surgery decreased an anti- inflammatory cytokine (IL10) and a stimulator of muscle growth (IGF-1), while it had no effect on a biomarker of oxidative stress (8- iso-PGF2α). In all subjects, there was a significant correlation between changes in IGF-1 following ACL surgery as compared to baseline (P < 0.05; R = 0.67)	II
Barker, 2009 [20]	Double-Blinded RCT	20	20 Male	18-45	Experimental: 28.9 ± 2.1 Placebo: 29.7 ± 2.0	Isokinetic testing, muscle biopsy, blood samples	Before and After	(n=10); 200 IU Vitamin E and 500 mg Vitamin C; 2x day	Placebo (n=9): 313 mg soybean oil, 110 mg gelatin, 10 mg water, 7.5 mg beeswax, 8.5 mg carob, 5 mg lecithin	14 weeks	At 90 min the vitamin group displayed a significant decrease in AA, an inverse correlation between AA and (interleukin) IL-8 (r2 = 0.50, P < 0.05), and a significantly lower IL-10 response than that of the PL group (P < 0.05).	II
Barker, 2015 [21]	Double-Blinded RCT	29	16 M, 13F	18-45	Experimental: 27.7 ± 1.0 Placebo: 29.2 ± 1.2	IKDC, WOMAC, Isokinetic testing, blood samples	Before and After	(n=15); 1200 mg Vitamin E 1000 mg Vitamin C; 2x daily	Placebo (n=14)	16 weeks	Serum cytokine concentrations, peak isometric force, and time to exhaustion during the exercise protocol were not significantly different between groups.	II
Wu, 2017 [22]	Double-Blinded RCT	13	13 Male	18-50	Not reported	Total body lean muscle mass, maximal extensor strength, KOOS	Before and After	(n=6) 200 mg intramuscular testosterone weekly	Placebo (N=7) Normal saline injection	26 weeks	There was an increase in mean lean mass at 6 weeks postoperatively in the testosterone group compared with a decrease of in the placebo group (*P* = .01). Extensor muscle strength of the uninjured leg also increased more from baseline in the testosterone group compared with the placebo group at 12 weeks (*P* = .04). There were no significant between-group differences in injured leg strength or clinical outcome scores.	I

ACL-R = Anterior Cruciate Ligament Reconstruction, M = Male, F = Female, VAS = Visual Analogue Scale, IKDC = International Knee Documentation Committee Subjective Form, LYS = Lysholm Knee Scoring Form, BMI = Body Mass Index, KOS = Knee Outcome Score, WOMAC = Western Ontario and McMaster Universities Osteoarthritis Index, KOOS = Knee Injury and Osteoarthritis Outcome Scores.

Table 2: Quality Assessment of Included Studies

Publication	Random Sequence Generation	Allocation Concealment	Selective Reporting	Other Bias	Blinding of Participants and Personnel	Blinding of Outcome Assessment	Incomplete Outcome Data	AHRQ Standards
Holm, 2005 [13]	low	low	low	unclear	low	low	low	fair
Vidriero, 2019 [14]	low	low	low	unclear	high	unclear	low	poor
Eraslan, 2014 [15]	high	high	low	unclear	low	low	low	poor
Laboute, 2013 [16]	low	unclear	low	unclear	low	low	low	fair
Tyler, 2004 [17]	low	low	low	unclear	low	low	low	fair
Barker, 2009 [18]	unclear	unclear	low	high	low	low	low	poor
Barker, 2009 [19]	unclear	unclear	low	high	low	low	low	poor
Barker, 2009 [20]	unclear	unclear	low	high	low	low	low	poor
Barker, 2015 [21]	low	unclear	low	unclear	low	low	low	fair
Wu, 2017 [22]	low	low	low	low	low	low	low	good

NOTE. Quality assessment of included studies with Cochrane Risk of Bias tool for Randomized Controlled Trials and conversion to Agency for Healthcare Research and Quality (AHRQ) Standards: “Good quality”: All criteria met (i.e., low for each domain); “Fair quality”: One criterion not met (i.e., high risk of bias for one domain) or 2 criteria unclear, and the assessment that this was unlikely to have

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biased the outcome, and there is no known important limitation that could invalidate the results; “Poor quality”: One criterion not met (i.e., high risk of bias for one domain) or 2 criteria unclear, and the assessment that this was likely to have biased the outcome, and there are important limitations that could invalidate the results; Poor quality: Two or more criteria listed as high or unclear risk of bias.

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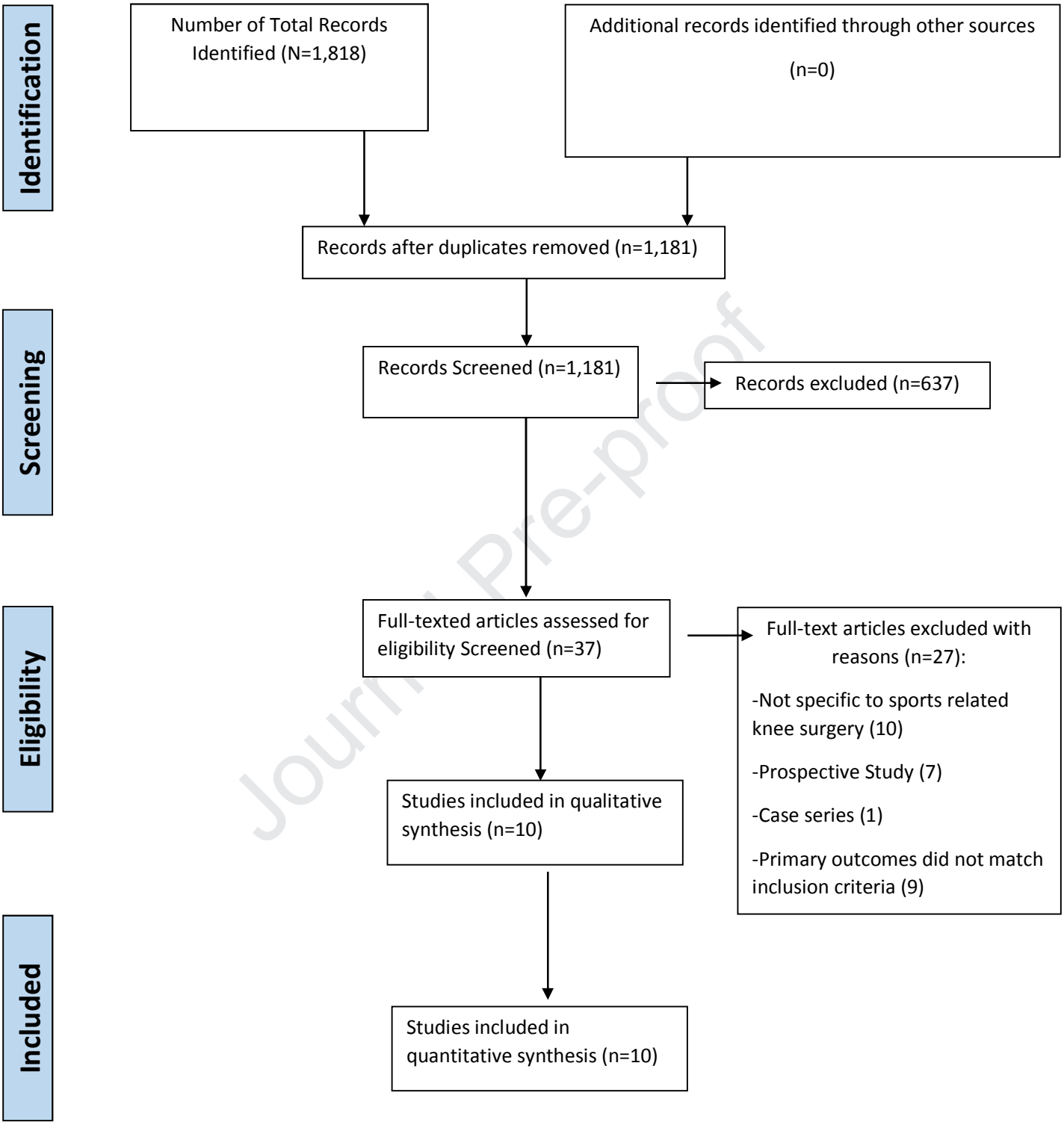
PRISMA Flow Chart



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